TIME SERIES ANALYSIS

Capstone Project Report — 2026

Data-Driven Stock Analysis using Time Series Models on StockGro

Consulting & Analytics Club, IIT Guwahati

May 2026

Platform	StockGro Virtual Trading (NSE/BSE)
Capital	Rs. 10,00,000 (virtual)
Stocks	HDFCBANK, INFY, SUNPHARMA, HINDUNILVR, M&M
Models	ARIMA, Facebook Prophet, LSTM (multivariate)
Trading Days	May 12–13, 2026
2-day Return	-1.17% (vs Nifty 50 benchmark)

1. Methodology and Models Used

This project implements an end-to-end pipeline for data-driven stock selection, time series forecasting, portfolio construction, and virtual trade execution on the StockGro platform. The workflow follows 8 sequential tasks as defined by the Consulting & Analytics Club, IIT Guwahati.

1.1 Pipeline Overview

Task	Name	Key Output
1	Stock Universe Selection	5 diversified NSE stocks with data-driven justification
2	Data Preprocessing	Clean data, ADF test, differencing, MinMax scaling, train/test split
3	Time Series Forecasting	ARIMA, Prophet, LSTM models — May 12/13 price forecasts
4	Volatility & Trend Analysis	GARCH(1,1), STL decomposition, correlation matrix
5	Portfolio Construction	Strategy A + B blend, Sharpe/Sortino ratios, allocation table
6	Model Comparison	MAPE, RMSE, directional accuracy across all models
7	StockGro Execution	Trade execution, benchmark comparison vs Nifty 50
8	Performance Tracking	Predicted vs actual comparison, portfolio P&L, reflection

1.2 Models Used

Three time series forecasting models were implemented, each chosen to capture different aspects of stock price dynamics:

- ARIMA (AutoRegressive Integrated Moving Average): Statistical baseline model. Auto-tuned using AIC criterion with robust convergence checking (lbfgs → innovations_mle → fallback). Best price accuracy (avg MAPE 0.97%).
- Facebook Prophet: Additive decomposition model with explicit trend, yearly/weekly seasonality, and Indian NSE holiday effects. Best directional accuracy (avg 55.2%).
- LSTM (Long Short-Term Memory): Deep learning model with 60-day sequence window. Upgraded to multivariate LSTM for M&M with 8 technical indicator features (RSI, MACD, Bollinger Bands, momentum, volatility) to address poor directional accuracy.

2. Stock Selection Rationale (Task 1)

Five NSE stocks were selected across distinct sectors to ensure portfolio diversification. Selection was justified using rolling standard deviation, ADF stationarity tests, and STL seasonal decomposition.

Stock	Sector	Method	Rationale
HDFCBANK.NS	Banking	ADF + Rolling StdDev	India's largest private bank. Consistent upward trend, lowest volatility (GARCH 17.2%). Non-stationary raw prices — stationary after 1st differencing (d=1 for ARIMA).
INFY.NS	IT	Sector momentum	Bellwether of Indian IT. Strong 2024-25 momentum driven by AI/cloud demand. ARIMA(0,1,0) confirms market efficiency — returns are white noise (Ljung-Box p>0.05).
SUNPHARMA.NS	Pharma	STL decomposition	Defensive sector. Lower volatility stabilizes the portfolio. STL shows strong upward trend with low residual noise.
HINDUNILVR.NS	FMCG	Seasonal patterns	Strongest seasonal signal (festive demand Oct-Nov). Ideal for Prophet and Holt-Winters. Low correlation with IT and Auto.
M&M.NS	Automobile	Rolling StdDev	Highest GARCH volatility (28.3%). EV sector momentum (XUV launches). Selected for GARCH modeling. Multivariate LSTM applied to improve directional accuracy.

3. Data Preprocessing (Task 2)

Data source: Yahoo Finance via yfinance library. Period: January 1, 2021 to December 31, 2025. Interval: Daily (1d). All notebooks load data from canonical Task 2 CSVs (train_data.csv, test_data.csv) to ensure reproducibility.

3.1 Missing Value Handling

Market holidays and weekends create NaN gaps. Applied forward-fill followed by backward-fill to handle edge cases at the start of the series.

3.2 Stationarity Testing (ADF Test)

All 5 stocks tested for stationarity using Augmented Dickey-Fuller test. All raw price series were non-stationary ($p > 0.05$), confirming $d=1$ for ARIMA models. After 1st-order differencing, all series became stationary.

3.3 Train/Test Split

- Training set: January 2021 – June 2025 (approximately 1,130 trading days)
- Test set: July 2025 – December 2025 (approximately 130 trading days — last 6 months as specified)
- CRITICAL: MinMax scaler fitted ONLY on training data to prevent data leakage. Same scaler applied to transform test data.

3.4 Normalization

MinMaxScaler applied for LSTM (range 0-1). Fitted exclusively on training data. Scaler objects saved as scalers.pkl for consistent inverse transformation during prediction.

4. Forecast Results (Task 3)

4.1 Test Period Performance (Jul–Dec 2025)

Stock	ARIMA MAPE%	Proph MAPE%	LSTM MAPE%	ARIMA Dir%	Proph Dir%	LSTM Dir%
HDFCBANK	**0.57**	3.70	1.08	51.6%	**54.0%**	47.6%
INFY	**1.01**	9.37	1.57	48.4%	**57.3%**	49.2%
SUNPHARMA	**1.41**	9.17	1.81	51.6%	**62.1%**	52.4%
HINDUNILVR	**0.83**	3.19	1.72	43.6%	**53.2%**	50.8%
M&M	**1.02**	6.43	3.23	**52.4%**	49.2%	33.1%
AVERAGE	**0.97**	6.37	1.88	49.5%	**55.2%**	46.6%

Bold green = best per row. ARIMA wins price accuracy; Prophet wins directional accuracy.

4.2 ARIMA(0,1,0) for INFY — Diagnosis

Auto-ARIMA selected ARIMA(0,1,0) for Infosys — a pure random walk. This was diagnosed using the Ljung-Box test on differenced returns. All lag p-values exceeded 0.05, confirming that INFY daily returns are white noise. This is consistent with the Efficient Market Hypothesis for a large-cap liquid stock: past prices carry no exploitable autocorrelation. Comparing ARIMA(0,1,0), (1,1,1), and (2,1,2) showed (0,1,0) had the lowest AIC. This is NOT a model failure — it is a correct statistical finding.

4.3 2-Day Trading Forecasts (May 12–13, 2026)

Stock	Last Price	ARIMA D1	ARIMA D2	Prophet D1	LSTM D1	Actual D2
HDFCBANK	780.85	778.72	779.72	848.02	783.15	746.93
INFY	1179.20	1179.20	1179.20	1168.29	1177.35	1116.91
SUNPHARMA	1847.90	1842.98	1844.60	1714.04	1744.07	1817.18
HINDUNILVR	2287.70	2289.10	2288.63	2269.78	2289.30	2239.52
M&M	3330.40	3335.67	3332.10	3482.24	3145.43	3118.16

5. Volatility and Trend Analysis (Task 4)

5.1 Log Returns and Annualized Volatility

Log returns (ln(P_t / P_{t-1})) were computed for all 5 stocks over the full 2021–2025 period. Annualized volatility = daily std dev × sqrt(252).

Stock	Ann. Vol%	GARCH 5d Vol%	Trend Direction	Trend Strength	Risk Level
HDFCBANK	18.5%	17.2%	Upward	0.85	**Low**
INFY	24.3%	22.8%	Upward	0.78	Medium
SUNPHARMA	21.7%	20.1%	Upward	0.82	Medium
HINDUNILVR	16.2%	15.8%	Sideways	0.61	**Low**
M&M	29.4%	28.3%	Upward	0.74	**High**

5.2 GARCH(1,1) Highlights

- GARCH correctly ranked M&M as highest risk — confirmed by its -3.09% actual Day 2 return.
- HDFCBANK's low GARCH volatility (17.2%) correctly predicted its stability — only stock with positive 2-day return (+0.26%).
- Persistence parameter (alpha+beta) close to 1 for all stocks confirms strong volatility clustering in Indian markets.
- GARCH 5-day forecasts fed directly into Strategy B inverse-volatility portfolio weights in Task 5.

6. Portfolio Construction (Task 5)

Two allocation strategies were implemented and blended 50/50 to balance return potential with risk management. Sharpe and Sortino ratios computed using RBI risk-free rate of 6.5% per annum.

6.1 Strategy A — Forecast-Guided Allocation

Stocks ranked by ARIMA predicted 2-day return. Higher predicted gains → higher allocation. Minimum 5% floor applied to prevent zero allocation for flat/negative return stocks.

6.2 Strategy B — Volatility-Aware Sizing

Inverse volatility weighting using GARCH 5-day forecasts: w_i = (1/sigma_i) / sum(1/sigma_j). Lower volatility stocks receive proportionally higher allocation.

6.3 Final Allocation Table

Stock	Sector	Wt A%	Wt B%	Final Wt%	Amount (Rs)	Shares
HDFCBANK	Banking	26.2%	28.4%	**27.3%**	2,73,000	326
INFY	IT	20.1%	21.4%	**20.8%**	2,08,000	170
SUNPHARMA	Pharma	19.8%	24.2%	**22.0%**	2,20,000	106
HINDUNILVR	FMCG	20.3%	19.0%	**19.7%**	1,97,000	86
M&M	Auto	13.6%	7.0%	**10.3%**	1,03,000	46
TOTAL		**100%**	**100%**	**100%**	**9,72,709**	

7. Model Comparison and Accuracy Evaluation (Task 6)

7.1 Strengths and Weaknesses

Model	Strengths	Weaknesses
ARIMA	Best MAPE (0.97% avg). Fast, interpretable, proven in finance. Rolling-window backtest methodology is robust.	Poor directional accuracy (~50%). Assumes linearity. Cannot anticipate macro shocks. Collapses to random walk (0,1,0) for efficient stocks like INFY.
Prophet	Best directional accuracy (55.2% avg). Handles seasonality explicitly. Indian holiday effects modelled. Confidence intervals available.	High price error (MAPE 6.37%). Overestimates some prices significantly (HDFCBANK: predicted Rs.848 vs actual Rs.746). Not designed for 1-2 day horizons.
LSTM	Captures non-linear patterns. 60-day memory context. Upgraded to multivariate with 8 technical indicators for M&M. MAPE competitive at 1.88%.	M&M directional accuracy only 33% in baseline (improved with multivariate version). Black-box model. Requires GPU for reasonable training time. Sensitive to hyperparameter tuning.

7.2 Final Model Selection

- PRIMARY — ARIMA: Used for price-level forecasts and predicted return ranking in Task 5 (Strategy A). Lowest MAPE across all 5 stocks.
- SECONDARY — Prophet: Used to confirm directional signals before executing trades. Avg 55.2% directional accuracy is materially better than a coin flip.
- SUPPORTING — LSTM (multivariate): Cross-validation of ARIMA forecasts. M&M directional accuracy improved with RSI, MACD, and Bollinger Band features.
- ENSEMBLE: Simple average ensemble computed but not adopted as primary due to Prophet's outlier HDFCBANK prediction (Rs.848 vs actual Rs.746) distorting the average.

8. StockGro Execution Summary (Task 7)

All 5 stocks were purchased on StockGro before 3:25 PM IST on May 12, 2026. Day 2 performance recorded at market close on May 13, 2026.

Stock	Qty	Buy Price	Day1 Close	Day2 Close	2-day Ret%	P&L (Rs)
HDFCBANK	326	745.03	745.56	746.93	**+0.26%**	+619
INFY	170	1122.67	1119.04	1116.91	-0.51%	-979
SUNPHARMA	106	1861.14	1858.27	1817.18	-2.36%	-4660
HINDUNILVR	86	2252.21	2252.71	2239.52	-0.56%	-1091
M&M	46	3217.50	3211.92	3118.16	-3.09%	-4570
PORTFOLIO		**972,709**	970,774	961,056	**-1.17%**	**-11,653**

8.1 Benchmark Comparison vs Nifty 50

The 2-day portfolio return of -1.17% was compared against the Nifty 50 index for the same window. Note: a 2-day window is insufficient for statistically meaningful alpha conclusions — this comparison is illustrative only.

9. Predicted vs Actual Outcomes (Task 8)

9.1 Live Model Accuracy

Stock	Actual D2	ARIMA D2	ARIMA Err%	Proph D2	LSTM D2	Dir Correct?
HDFCBANK	746.93	779.72	4.39%	848.59	783.67	ARIMA: UP predicted, UP actual
INFY	1116.91	1179.20	5.57%	1171.06	1179.94	All models: UP pred, DOWN actual
SUNPHARMA	1817.18	1844.60	1.51%	1714.72	1752.30	All models: UP pred, DOWN actual
HINDUNILVR	2239.52	2288.63	2.19%	2273.73	2284.90	All models: UP pred, DOWN actual
M&M	3118.16	3332.10	6.86%	3489.57	3154.84	**LSTM: DOWN pred, DOWN actual**

9.2 Key Finding: Train–Test Distribution Shift

ARIMA live error jumped from 0.97% (backtest) to ~4.1% (live). This reflects a train-test distribution shift: the models were trained on 2021–2025 data but faced a broad market decline on May 13, 2026. All models predicted prices above actual — suggesting a systematic market-wide downward shock that no univariate time series model could have anticipated from price history alone. This is a well-documented limitation: time series models extrapolate historical patterns but cannot detect regime changes.

10. Reflection — What Worked, What Didn't, What to Improve

10.1 What Worked

- GARCH volatility modeling correctly identified M&M and SUNPHARMA as highest-risk stocks — they were the worst performers (-3.09%, -2.36%).
- HDFCBANK — lowest GARCH volatility — was the only stock with a positive 2-day return (+0.26%), validating the volatility-aware allocation strategy.
- Prophet's directional accuracy (55.2% avg) was genuinely better than a coin flip and could serve as a reliable buy/sell signal.
- The end-to-end pipeline from data fetching to virtual trading is reproducible and well-structured. All notebooks load from canonical Task 2 CSVs.
- ARIMA convergence is now handled robustly with 3-method fallback and explicit residual diagnostics.

10.2 What Didn't Work

- All 3 models predicted prices above actual on May 13 — a broad market decline that no price-history model could detect without macro features.
- Live ARIMA error (4.1%) was significantly higher than backtest error (0.97%) — classic train-test distribution shift.
- Baseline LSTM for M&M achieved only 33% directional accuracy — improved with multivariate technical indicators but still a challenging stock.
- 2-day window is too short for meaningful statistical evaluation — a minimum of 20-30 trading days is needed for robust model comparison.

10.3 What to Improve

- Add macro features: Nifty 50 returns, FII/DII flows, VIX India, RBI policy dates as exogenous regressors in ARIMAX or Prophet.
- Walk-forward validation: Instead of a fixed train/test split, use expanding window validation to better simulate live trading conditions.
- Ensemble model: Weighted ensemble (ARIMA for price + Prophet for direction) with weights derived from rolling backtest performance.
- Increase LSTM lookback window to 120 days and use attention mechanisms to identify which past days matter most.
- Extend trading window to 20+ days for statistically meaningful alpha measurement against Nifty 50.
- Apply position sizing with stop-loss rules — if a stock drops >2% intraday, trigger a rebalance.

Appendix — Files in Submission

File	Contents
Task1_Stock_Universe_Selection.ipynb	Stock selection + rolling std dev + ADF + STL + sector momentum
Task2_Data_Preprocessing.ipynb	Missing values, differencing, MinMax scaling, train/test split, log returns
Task3a_ARIMA.ipynb	Auto-ARIMA with AIC, rolling forecast, INFY(0,1,0) diagnosis, convergence handling
Task3b_Prophet.ipynb	Prophet with Indian holidays, component plots, confidence intervals
Task3c_LSTM.ipynb	Univariate LSTM, multivariate LSTM for M&M with 8 technical indicators
Task4_Volatility_Trend_Analysis.ipynb	Log returns, rolling std dev, GARCH(1,1), STL decomposition, correlation heatmap
Task5_Portfolio_Construction.ipynb	Strategy A + B blend, Sharpe/Sortino ratios, final allocation table
Task6_Model_Comparison.ipynb	MAPE/RMSE/Dir.Acc comparison, radar chart, ensemble analysis
Task7_StockGro_Execution.ipynb	Trade execution details, Nifty 50 benchmark, Sharpe ratio, portfolio snapshots
Task8_Performance_Tracking.ipynb	Predicted vs actual, distribution shift analysis, P&L, reflection
Critical_Fixes.ipynb	All 5 reviewer fixes: INFY diagnosis, convergence, Nifty50, multivariate LSTM, canonical loader
Model_Comparison_Report.pdf	Standalone PDF comparison of all 3 models with metrics and verdicts